Chapter 1

Demo problem: Flow past a cylinder with a waving flag

In this example we consider the flow in a 2D channel that contains a cylinder with a waving "flag". This is a "warm-up problem" for the solution of Turek & Hron's FSI benchmark problem discussed in another tutorial.

1.1 The Problem

The figure below shows a sketch of the problem: A 2D channel of height H^* and length L^* contains a cylinder of diameter d^*, centred at (X_c^*, Y_c^*) to which a "flag" of thickness H_{flag}^* and length L_{flag}^* is attached. We assume that the flag performs time-periodic oscillations with period T^*. Steady Poiseuille flow with average velocity U^* is imposed at the left end of the channel while we assume the outflow to be parallel and axially traction-free.

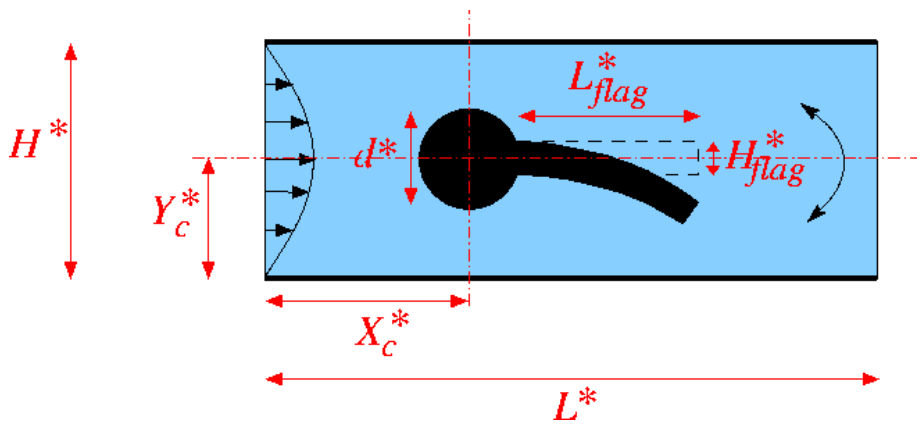

Figure 1.1 Sketch of the problem in dimensional variables.

We non-dimensionalise all length and coordinates on the diameter of the cylinder, d^*, time on the natural timescale of the flow, d^*/U^*, the velocities on the mean velocity, U^*, and the pressure on the viscous scale. The problem is then governed by the non-dimensional Navier-Stokes equations

$$Re\left(St\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right],$$

where $Re = \rho U^* H_0^*/\mu$ and $St = 1$, and

$$\frac{\partial u_i}{\partial x_i} = 0,$$

subject to parabolic inflow

$$\mathbf{u} = 6x_2(1 - x_2)\mathbf{e}_1$$

at the inflow cross-section; parallel, axially-traction-free outflow at the outlet; and no-slip on the stationary channel walls and the surface of the cylinder, $\mathbf{u} = \mathbf{0}$. The no-slip condition on the moving flag is

$$\mathbf{u} = \frac{\partial \mathbf{R}_w(\xi_{[top,tip,bottom]}, t)}{\partial t}$$

where $\xi_{[top,tip,bottom]}$ are Lagrangian coordinates parametrising the three faces of the flag. The flag performs oscillations with non-dimensional period $T = T^* U^* / H^*_{tot}$. Here is a sketch of the non-dimensional version of the problem:

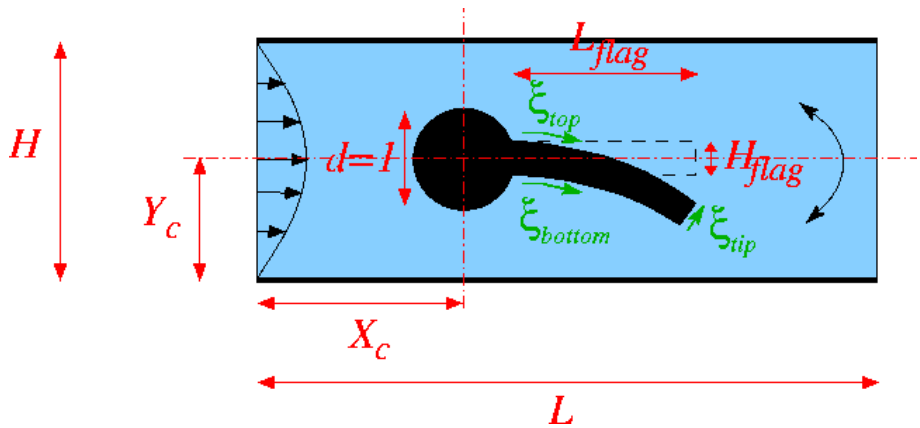

Figure 1.2 Sketch of the problem in dimensionless variables, showing the Lagrangian coordinates that parametrise the three faces of the flag.

1.2 Results

The figure below shows a snapshot of the flow field
(pressure contours and instantaneous streamlines) for a Reynolds number of $Re = 100$ and an oscillation period of $T = 10$, as well as the corresponding fluid mesh. Note how `oomph-lib's` automatic mesh adaptation has refined the mesh in the high-shear regions near the front of the cylinder and at the trailing edge of the flag.

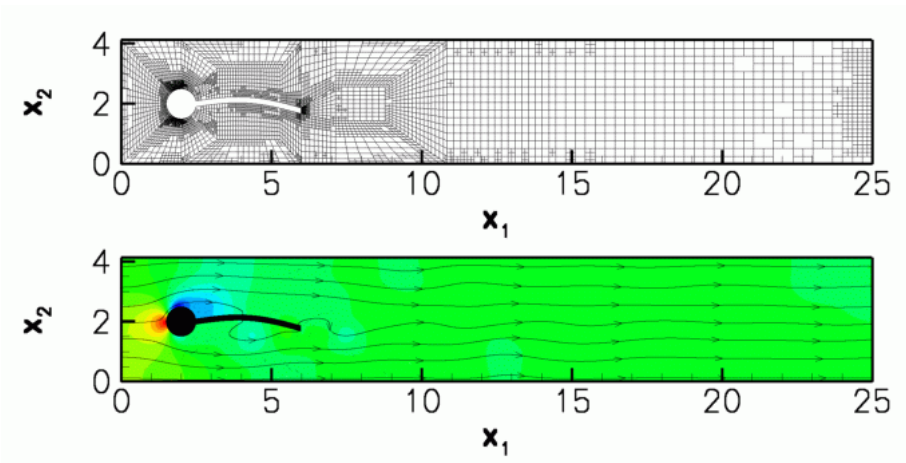

Figure 1.3 Mesh (top) and flow field (bottom).

The corresponding `animation` illustrates the algebraic node update strategy (implemented with an `AlgebraicMesh`, discussed in more detail in `another tutorial`) and the evolution of the flow field. Note that the instantaneous streamlines intersect the (impermeable) flag because the flag is not stationary.

1.3 The global parameters

As usual we use a namespace to define the (single) global parameter, the Reynolds number.

```
//====start_of_global_parameters==========================================
/// Global parameters
//========================================================================
namespace Global_Parameters
{
 /// Reynolds number
 double Re=100.0;
}
```

1.4 Specification of the flag geometry

We specify the flag geometry and its time-dependent motion by representing its three faces by `GeomObjects`, each parametrised by its own Lagrangian coordinate, as shown in the sketch above. The geometry of the cylinder is represented by one of `oomph-lib's` generic `GeomObjects`, the `Circle`.
We enclose the relevant data in its own namespace and start by defining the period of the oscillation, and the geometric parameters controlling the flag's initial shape and its subsequent motion.

```
//====start_of_flag_definition============================================
/// Namespace for definition of flag boundaries
//========================================================================
namespace Flag_definition
{

 /// Period of prescribed flag oscillation
 double Period=10.0;

 /// Height of flag
 double H=0.2;

 /// Length of flag
 double L=3.5;

 /// x position of centre of cylinder
 double Centre_x=2.0;

 /// y position of centre of cylinder
 double Centre_y=2.0;

 /// Radius of cylinder
 double Radius=0.5;

 /// Amplitude of tip deflection
 double Amplitude=0.33;

 /// Pointer to the global time object
 Time* Time_pt=0;
```

We choose the motion of the flag such that it vaguely resembles that expected in the corresponding FSI problem: We subject the flag's upper and lower faces to purely vertical displacements that deform them into a fraction of a sine wave, while keeping the face at the tip of the flag straight and vertical. We implement this by prescribing the time-dependent motion of the two vertices at the tip of the flag with two functions:

```
 /// Time-dependent vector to upper tip of the "flag"
 Vector<double> upper_tip(const double& t)
  {
   double tmp_ampl=Amplitude;
   if (t<=0.0) tmp_ampl=0.0;
   Vector<double> uppertip(2);
   uppertip[0]= Centre_x+Radius*sqrt(1.0-H*H/(4.0*Radius*Radius))+L;
   uppertip[1]= Centre_y+0.5*H-
    tmp_ampl*sin(2.0*MathematicalConstants::Pi*t/Period);
   return uppertip;
  }

 /// Time-dependent vector to bottom tip of the "flag"
 Vector<double> lower_tip(const double& t)
  {
   double tmp_ampl=Amplitude;
   if (t<=0.0) tmp_ampl=0.0;
   Vector<double> lowertip(2);
   lowertip[0]= Centre_x+Radius*sqrt(1.0-H*H/(4.0*Radius*Radius))+L;
   lowertip[1]= Centre_y-0.5*H-
    tmp_ampl*sin(2.0*MathematicalConstants::Pi*t/Period);
   return lowertip;
  } // end of bottom tip
```

This information is then used in three custom-written `GeomObject` that define the time-dependent shape of the

three faces. Here is the one describing the shape of the upper face:

```
//-----start_of_top_of_flag-------------------------------------
/// GeomObject that defines the upper boundary of the flag
//---------------------------------------------------------------
 class TopOfFlag : public GeomObject
 {

 public:

  /// Constructor: It's a 1D object in 2D space
  TopOfFlag() : GeomObject(1,2) {}

  /// Destructor (empty)
  ~TopOfFlag(){}

  /// Return the position along the top of the flag (xi[0] varies
  /// between 0 and Lx)
  void position(const unsigned& t,const Vector<double> &xi, Vector<double> &r)
   const
   {
    // Compute position of fixed point on the cylinder
    Vector<double> point_on_circle(2);
    point_on_circle[0]=Centre_x+Radius*sqrt(1.0-H*H/(4.0*Radius*Radius));
    point_on_circle[1]=Centre_y+H/2.0;

    r[0] = point_on_circle[0]+xi[0]/L*(upper_tip(Time_pt->time(t))[0]-
                                       point_on_circle[0]);

    r[1] = point_on_circle[1]+xi[0]/L*(upper_tip(Time_pt->time(t))[1]-
                                       point_on_circle[1])+
     1.0/3.0*sin((r[0]-point_on_circle[0])/
                 (upper_tip(Time_pt->time(t))[0]-
                  point_on_circle[0])*MathematicalConstants::Pi)
     *sin(2.0* MathematicalConstants::Pi*Time_pt->time(t)/Period);

   }

  /// Current position
  void position(const Vector<double> &xi, Vector<double> &r) const
   {
    return position(0,xi,r);
   }

  /// Number of geometric Data in GeomObject: None.
  unsigned ngeom_data() const {return 0;}

};
```

[We omit the definition of the other two `GeomObjects`, `BottomOfFlag` and `TipOfFlag` in in the interest of brevity. Their definitions can be found in the source code.] We provide storage for the pointers to the four `GeomObjects` required for the representation of the flag,

```
 /// Pointer to GeomObject that bounds the upper edge of the flag
 TopOfFlag* Top_flag_pt=0;

 /// Pointer to GeomObject that bounds the bottom edge of the flag
 BottomOfFlag* Bottom_flag_pt=0;

 /// Pointer to GeomObject that bounds the tip edge of the flag
 TipOfFlag* Tip_flag_pt=0;

 /// Pointer to GeomObject of type Circle that defines the
 /// central cylinder.
 Circle* Cylinder_pt=0;
```

and provide a setup function that generates the `GeomObjects` and stores the pointer to the global `Time` object that will be created by the `Problem`:

```
 /// Create all GeomObjects needed to define the cylinder and the flag
 void setup(Time* time_pt)
 {
  // Assign pointer to global time object
  Time_pt=time_pt;

  // Create GeomObject of type Circle that defines the
  // central cylinder.
  Cylinder_pt=new Circle(Centre_x,Centre_y,Radius);

  /// Create GeomObject that bounds the upper edge of the flag
  Top_flag_pt=new TopOfFlag;

  /// Create GeomObject that bounds the bottom edge of the flag
  Bottom_flag_pt=new BottomOfFlag;
```

```
  /// Create GeomObject that bounds the tip edge of the flag
  Tip_flag_pt=new TipOfFlag;
 }

} // end of namespace that specifies the flag
```

1.5 The mesh

The figure below shows a sketch of the (unrefined) mesh and the enumeration of its boundaries. Various different implementations of the mesh exist, allowing the user to perform the node-update in response to the motion of the flag by a `Domain/MacroElement` - based procedure, or by using an algebraic node update. In either case, only the nodes in the elements that are shaded in yellow (or refined elements that are generated from these) participate in the node-update.

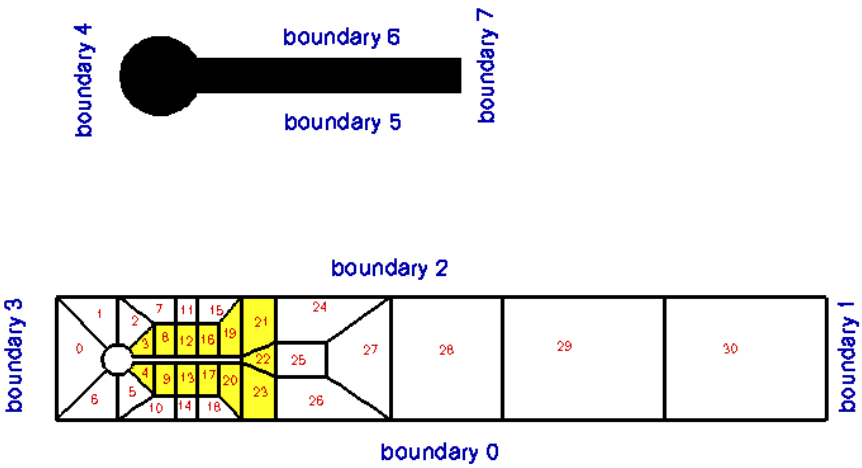

Figure 1.4 The (unrefined) mesh. Only nodes in the yellow regions participate in the node-update.

The node update strategy is illustrated in the animation of the flow field and the mesh motion.

1.6 The driver code

The driver code has the usual structure for a time-dependent problem: We store the command line arguments, create a `DocInfo` object, and assign the parameters that specify the dimensions of the channel.

```
//======start_of_main==================================================
/// Driver code -- pass a command line argument if you want to run
/// the code in validation mode where it only performs a few steps
//=====================================================================
int main(int argc, char* argv[])
{

 // Store command line arguments
 CommandLineArgs::setup(argc,argv);
 // Set up doc info
 DocInfo doc_info;
 doc_info.set_directory("RESLT");
 doc_info.number()=0;
 // Length and height of domain
 double length=25.0;
 double height=4.1;
```

We build the problem using a mesh in which the node update is performed by the `AlgebraicMesh` class. (The driver code also provides the option to use a `Domain/MacroElement` - based node update – this option is chosen via suitable #ifdefs; see the source code for details).

```
 // Create Problem with AlgebraicMesh-based node update
 TurekNonFSIProblem
  <AlgebraicElement<RefineableQCrouzeixRaviartElement<2> > >
  problem(length, height);
```

Next we set up the time-stepping (as usual, fewer timesteps are performed during a validation run which is identified by a non-zero number of command line arguments):

```
// Number of timesteps per period
 unsigned nsteps_per_period=40;

 // Number of periods
```

```
 unsigned nperiod=3;

 // Number of timesteps  (reduced for validation)
 unsigned nstep=nsteps_per_period*nperiod;
 if (CommandLineArgs::Argc>1)
  {
   nstep=2;
   // Also reduce the Reynolds number to reduce the mesh refinement
   Global_Parameters::Re=10.0;
  }

 //Timestep:
 double dt=Flag_definition::Period/double(nsteps_per_period);

 /// Initialise timestep
 problem.initialise_dt(dt);
```

We start the simulation with a steady solve, allowing up to three levels of adaptive refinement (fewer if we are performing a validation run):

```
 // Solve adaptively with up to max_adapt rounds of refinement (fewer if
 // run during self-test)
 unsigned max_adapt=3;
 if (CommandLineArgs::Argc>1)
  {
   max_adapt=1;
  }

 /// Output intial guess for steady Newton solve
 problem.doc_solution(doc_info);
 doc_info.number()++;
```

Finally, we enter the proper timestepping loop, allowing one spatial adaptation per timestep and suppressing the re-assignment of initial conditions following an adaptation by setting the parameter `first` to false. (See the discussion of timestepping with automatic mesh adaptation in `another tutorial`.)

```
 // Do steady solve first -- this also sets the history values
 // to those corresponding to an impulsive start from the
 // steady solution
 problem.steady_newton_solve(max_adapt);

 /// Output steady solution = initial condition for subsequent unsteady solve
 problem.doc_solution(doc_info);
 doc_info.number()++;

 /// Reduce the max number of adaptations for time-dependent simulation
 max_adapt=1;

 // We don't want to re-assign the initial condition after the mesh
 // adaptation
 bool first=false;
// Timestepping loop
 for (unsigned istep=0;istep<nstep;istep++)
  {
   // Solve the problem
   problem.unsteady_newton_solve(dt,max_adapt,first);

   // Output the solution
   problem.doc_solution(doc_info);

   // Step number
   doc_info.number()++;
  }
}//end_of_main
```

1.7 The Problem class

The problem class provides an access function to the mesh (note the use of #ifdefs to choose the correct one), and defines the interfaces for the usual member functions, either empty or explained in more detail below.

```
//======= start_of_problem_class=====================================
/// Flow around a cylinder with flag
//====================================================================
template<class ELEMENT>
class TurekNonFSIProblem : public Problem
{

public:

 /// Constructor: Pass length and height of domain.
 TurekNonFSIProblem(const double &length,
                    const double &height);

 /// Update the problem specs after solve (empty)
 void actions_after_newton_solve() {}
```

```
 /// Update the problem specs before solve (empty)
 void actions_before_newton_solve() {}

 /// After adaptation: Unpin pressures and pin redudant pressure dofs.
 void actions_after_adapt();

 /// Update the velocity boundary condition on the flag
 void actions_before_implicit_timestep();


#ifdef USE_MACRO_ELEMENTS

 /// Access function for the specific mesh
 RefineableCylinderWithFlagMesh<ELEMENT>* mesh_pt()
  {
   return dynamic_cast<RefineableCylinderWithFlagMesh<ELEMENT>*>
    (Problem::mesh_pt());
  }

#else

 /// Access function for the specific mesh
 RefineableAlgebraicCylinderWithFlagMesh<ELEMENT>* mesh_pt()
  {
   return dynamic_cast<RefineableAlgebraicCylinderWithFlagMesh<ELEMENT>*>
    (Problem::mesh_pt());
  }
#endif

 /// Doc the solution
 void doc_solution(DocInfo& doc_info);


private:

 /// Height of channel
 double Height;

};//end_of_problem_class
```

1.8 The problem constructor

The initial assignment of zero velocity and pressure provides a very poor initial guess for the preliminary steady Newton solve. We therefore increase the maximum residuals and iteration counts allowed in the Newton iteration before creating the timestepper and setting up the `GeomObjects` required to parametrise the flag:

```
//=====start_of_constructor=================================================
/// Constructor: Pass length and height of domain.
//==========================================================================
template<class ELEMENT>
TurekNonFSIProblem<ELEMENT>::TurekNonFSIProblem(
 const double &length, const double &height) : Height(height)
{

 // Bump up Newton solver parameters to allow for crappy initial guesses
 Max_residuals=100.0;;
 Max_newton_iterations=50;

 // Allocate the timestepper
 add_time_stepper_pt(new BDF<2>);

 // Setup flag/cylinder geometry
 Flag_definition::setup(time_pt());
```

Next we build the mesh, passing the pointers to the various `GeomObjects` and the geometric parameters to its constructor.

```
 // Build mesh (with AlgebraicMesh-based node update)
 Problem::mesh_pt()=new RefineableAlgebraicCylinderWithFlagMesh<ELEMENT>
  (Flag_definition::Cylinder_pt,
   Flag_definition::Top_flag_pt,
   Flag_definition::Bottom_flag_pt,
   Flag_definition::Tip_flag_pt,
   length, height,
   Flag_definition::L,
   Flag_definition::H,
   Flag_definition::Centre_x,
   Flag_definition::Centre_y,
   Flag_definition::Radius,
   time_stepper_pt());
```

We perform two rounds of uniform refinement (the Newton solver does not converge on coarser meshes) before creating an error estimator for the subsequent automatic mesh adaptation.

```
 // Refine uniformly
 mesh_pt()->refine_uniformly();
```

```
mesh_pt()->refine_uniformly();

// Set error estimator
Z2ErrorEstimator* error_estimator_pt=new Z2ErrorEstimator;
mesh_pt()->spatial_error_estimator_pt()=error_estimator_pt;
```

Both velocity components are imposed by Dirichlet conditions (either via a no-slip condition or via the imposed inflow profile) on all boundaries, apart from the outflow cross-section (mesh boundary 1), where the axial velocity is unknown.

```
// Set the boundary conditions for this problem: All nodes are
// free by default -- just pin the ones that have Dirichlet conditions
// here.
//Pin both velocities at all boundaries apart from outflow
unsigned num_bound = mesh_pt()->nboundary();
for(unsigned ibound=0;ibound<num_bound;ibound++)
 {
  unsigned num_nod= mesh_pt()->nboundary_node(ibound);
  for (unsigned inod=0;inod<num_nod;inod++)
   {
    // Parallel, axially traction free outflow at downstream end
    if (ibound != 1)
     {
      mesh_pt()->boundary_node_pt(ibound,inod)->pin(0);
      mesh_pt()->boundary_node_pt(ibound,inod)->pin(1);
     }
    else
     {
      mesh_pt()->boundary_node_pt(ibound,inod)->pin(1);
     }
   }
 } // done bc
```

Next we complete the build of the elements, passing the relevant pointers to physical parameters,

```
 // Pass pointer to Reynolds number to elements
 unsigned nelem=mesh_pt()->nelement();
 for (unsigned e=0;e<nelem;e++)
  {
   // Upcast from GeneralisedElement to the present element
   ELEMENT* el_pt = dynamic_cast<ELEMENT*>(mesh_pt()->element_pt(e));

   //Set the Reynolds number
   el_pt->re_pt() = &Global_Parameters::Re;

   //Set the Womersley number (assuming St=1)
   el_pt->re_st_pt() = &Global_Parameters::Re;
  }
```

and impose the steady Poiseuille profile at the inlet (mesh boundary 3).

```
 // Setup Poiseuille flow along boundary 3
 unsigned ibound=3;
 unsigned num_nod= mesh_pt()->nboundary_node(ibound);
 for (unsigned inod=0;inod<num_nod;inod++)
  {
   double ycoord = mesh_pt()->boundary_node_pt(ibound,inod)->x(1);

   // Set Poiseuille velocity
   double uy = 6.0*ycoord/Height*(1.0-ycoord/Height);

   mesh_pt()->boundary_node_pt(ibound,inod)->set_value(0,uy);
   mesh_pt()->boundary_node_pt(ibound,inod)->set_value(1,0.0);
  }
```

Finally, we pin the redundant pressure degrees of freedom (see `another tutorial` for details), and assign the equations numbers.

```
 // Pin redudant pressure dofs
 RefineableNavierStokesEquations<2>::
  pin_redundant_nodal_pressures(mesh_pt()->element_pt());

 // Assign equations numbers
 cout «"Number of equations: " « assign_eqn_numbers() « endl;
} // end of constructor
```

1.9 Actions before adapt

After each adaptation, we unpin and re-pin all redundant pressures degrees of freedom (see `another tutorial` for details). Since the inflow profile is parabolic, it is interpolated correctly from "father" to "son" elements during mesh refinement so no further action is required.

```
//=====actions_after_adapt================================================
/// Actions after adapt
//=========================================================================
template <class ELEMENT>
void TurekNonFSIProblem<ELEMENT>::actions_after_adapt()
{
 // Unpin all pressure dofs
 RefineableNavierStokesEquations<2>::
  unpin_all_pressure_dofs(mesh_pt()->element_pt());
 // Pin redundant pressure dofs
 RefineableNavierStokesEquations<2>::
  pin_redundant_nodal_pressures(mesh_pt()->element_pt());
} // end_of_actions_after_adapt
```

1.10 Actions before the timestep

Before each timestep we update the nodal positions in the mesh and re-apply the no-slip condition at the nodes on mesh boundaries 5 - 7.

```
//==== start_of_actions_before_implicit_timestep==========================
/// Actions before implicit timestep: Update velocity boundary conditions
//=========================================================================
template <class ELEMENT>
void TurekNonFSIProblem<ELEMENT>::actions_before_implicit_timestep()
{

 // Update the domain shape
 mesh_pt()->node_update();

 // Moving leaflet: No slip; this implies that the velocity needs
 // to be updated in response to flag motion
 for( unsigned ibound=5;ibound<8;ibound++)
  {
   unsigned num_nod=mesh_pt()->nboundary_node(ibound);
   for (unsigned inod=0;inod<num_nod;inod++)
    {
     // Which node are we dealing with?
     Node* node_pt=mesh_pt()->boundary_node_pt(ibound,inod);

     // Apply no slip
     FSI_functions::apply_no_slip_on_moving_wall(node_pt);
    }
  }
} //end_of_actions_before_implicit_timestep
```

1.11 Post Processing

The function doc_solution(...) documents the results.

```
//==start_of_doc_solution=================================================
/// Doc the solution
//=========================================================================
template<class ELEMENT>
void TurekNonFSIProblem<ELEMENT>::doc_solution(DocInfo& doc_info)
{

 ofstream some_file;
 char filename[100];

 // Number of plot points
 unsigned npts;
 npts=5;

 // Output solution
 snprintf(filename, sizeof(filename), "%s/soln%i.dat",doc_info.directory().c_str(),
         doc_info.number());
 some_file.open(filename);
 mesh_pt()->output(some_file,npts);
 some_file.close();

} // end_of_doc_solution
```

1.12 Comments and Exercises

- Compare the results obtained when the node update is performed with the algebraic or the `Macro↩ Element/Domain-based` node update. (`oomph-lib's` build machinery will automatically generate both versions of the code, using the `-DUSE_MACRO_ELEMENTS` compilation flag).

1.13 Acknowledgements

- This code was originally developed by Stefan Kollmannsberger and his students Iason Papaioannou and Orkun Oezkan Doenmez. It was completed by Floraine Cordier.

1.14 Source files for this tutorial

- The source files for this tutorial are located in the directory:

  `demo_drivers/navier_stokes/turek_flag_non_fsi/`

- The driver code is:

  `demo_drivers/navier_stokes/turek_flag_non_fsi/turek_flag_non_fsi.cc`

1.15 PDF file

A `pdf version` of this document is available. \